Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/05/2018
PI Name:	Barger, Laura Ph.D.		
Project Title:	The Impact of Objectively Measured Sleep Deficiency and Circadian Misalignment on Performance During Spaceflight		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02115-5817	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Crew Health (FLAGSHIP, NSBRI, OMNIBUS). Appendix A-Crew Health, Appendix B-NSBRI, Appendix C-Omnibus
Start Date:	06/01/2016	**End Date:**	05/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/18/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Basner, Mathias M.D., Ph.D. (University of Pennsylvania) Czeisler, Charles M.D., Ph.D. (Brigham And Women's Hospital) Dinges, David Ph.D. (University of Pennsylvania) Seaton, Kim Ph.D. (KBR Wyle/NASA Johnson Space Center) Flynn-Evans, Erin Ph.D. (NASA Ames Research Center) Wang, Wei Ph.D. (Brigham And Women's Hospital)		
Grant/Contract No.:	NCC 9-58-HFP04504		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Alertness and performance, which are critical to the success of spaceflight operations, are governed by: 1) circadian phase (biological time of day); 2) number of hours awake (acute sleep deficiency); and 3) nightly sleep duration (chronic sleep deficiency). In ground-based studies, each of these factors has been independently associated with decrements in neurobehavioral performance and an increased risk of errors and accidents. A ten-year study of objectively measured sleep during spaceflight revealed that sleep deficiency and use of sleep-promoting medication was pervasive during both short and long duration spaceflight. On 20% of nights aboard the International Space Station (ISS), crewmembers experienced circadian misalignment. Although the decade-long study of sleep during spaceflight did not collect any performance data, crewmembers completed the Spaceflight Cognitive Assessment Tool for Windows (WinSCAT) and Psychomotor Vigilance Task (PVT) Reaction Self-Test (RST) over the same time period. To determine the association among circadian phase, acute sleep deficiency, and chronic sleep deficiency, we integrated and analyzed the existing sleep and performance databases. Paired t-tests and linear mixed-effects models will be used to test the hypotheses that during spaceflight, (1) on days when crewmembers are estimated to have circadian misalignment, they will have greater performance decrements (e.g., increased response time) than on days when they are estimated to be in circadian alignment; (2) crewmembers with objectively measured acute sleep deficiency (e.g., awake for more than 16 hours) will have greater performance decrements than crewmembers who have been awake less than 10 hours; (3) crewmembers with objectively measured chronic sleep deficiency (i.e., < 6 hours of sleep, on average, for the 4 preceding nights) will have greater performance decrements than crewmembers who slept more than 6 hours per night, on average, for the 4 preceding nights. Performance was consistently impaired on the Reaction Self-Test with acute and chronic sleep deficiency, including increased reaction time and increased lapses. The smaller WinSCAT dataset revealed less conclusive results, likely due to improvements in performance over the duration of the spaceflight mission and the option of not taking the assessment when fatigued. Similar to ground-based studies, performance is impaired with acute and chronic sleep deficiency. Further research is required to determine the extent of the impairment and relationship to operational task performance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We investigated whether extended wakefulness, chronic sleep deficiency, and circadian misalignment were associated with performance decrements during spaceflight. The results of this research have major implications for those who work extended-duration shifts, the one-third of American workers who report chronic sleep deficiency, and the 15% of the labor force employed as shift workers. The results of this work could lead to improved policies and/or fatigue countermeasures, thereby improving the health, productivity and safety not only of astronauts during spaceflight, but also those employed here on Earth.
Task Progress:	This one-year research project required the merging of three different databases, two from the Life Sciences Data Archive (LSDA) and one from Lifetime Surveillance of Astronaut Health (LSAH). We made a presentation to the LSAH and LSDA advisory boards and requested the advisory boards approve the release of attributable Reaction Self-Test data from the LSDA Repository and WinSCAT data from the LSAH Repository for comparison with sleep log and actigraphy data (i.e., data we collected and housed in LSDA) to objectively measure the impact of sleep deficiency and circadian misalignment on performance during spaceflight. The request to use data from the sleep and performance databases was approved by advisory boards. An amendment to perform these secondary analyses was submitted and approved at the NASA Johnson Space Center Institutional Review Board. The LSDA reviewed the consents given by all the crewmembers in the three databases. The data from crewmembers which chose free use of any of their data or free use of the sleep and performance data were released to us. The three databases were aligned by date and time so that comparisons between sleep and performance could be analyzed. Statistical analysis was performed by the statistician at Brigham and Women's Hospital. A web-based presentation was made to co-investigators at University of Pennsylvania and University of Texas Medical Branch. Following the submission of this final report, a presentation will be made to National Space Biomedical Research Institute (NSBRI) personnel and a manuscript will be submitted to a scientific journal for publication.
Bibliography Type:	Description: (Last Updated: 06/21/2024)
Abstracts for Journals and Proceedings	Barger LK, Sullivan JP, Flynn-Evans EE, Basner M, DInges DF, Seaton KA, Wang W, Czeisler CA. "The impact of objectively measured sleep deficiency and circadian misalignment on performance during spaceflight." 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017
Articles in Peer-reviewed Journals	Grant LK, Kent BA, Rahman SA, St Hilaire MA, Kirkley CL, Gregory KB, Clark T, Hanifin JP, Barger LK, Czeisler CA, Brainard GC, Lockley SW, Flynn-Evans EE. "The effect of a dynamic lighting schedule on neurobehavioral performance during a 45-day simulated space mission." Sleep Adv. 2024 May 30;5(1):zpae032. https://doi.org/10.1093/sleepadvances/zpae032 ; PMID: 38903700; PMCID: PMC11187988 , May-2024